

## ► BIOCHEMISTRY—MOLECULAR

**Chromatin structure**

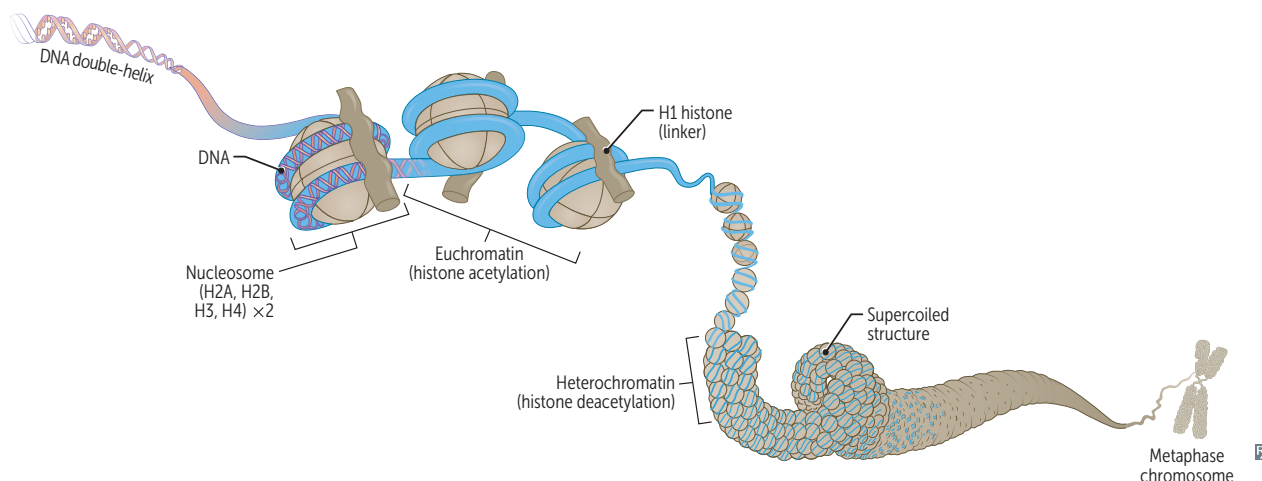
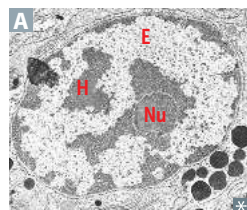
DNA exists in the condensed, chromatin form to fit into the nucleus. DNA loops twice around a histone octamer to form a nucleosome (“beads on a string”). H1 binds to the nucleosome and to “linker DNA,” thereby stabilizing the chromatin fiber.

DNA has  $\ominus$  charge from phosphate groups.

Histones are **large** and have  $\oplus$  charge from **lysine** and **arginine**.

DNA and histone synthesis occurs during S phase.

Mitochondria have their own DNA, which is circular and does not bind histones.

**Heterochromatin**

Condensed, appears darker on EM (labeled H in **A**; Nu, nucleolus). Sterically inaccessible, thus transcriptionally inactive.  $\uparrow$  methylation,  $\downarrow$  acetylation.

**Heterochromatin** = **highly condensed** (**hidden chromatin**).

Barr bodies (inactive X chromosomes) may be visible on the periphery of nucleus.

**Euchromatin**

Less condensed, appears lighter on EM (labeled E in **A**). Transcriptionally active, sterically accessible.

*Eu* = true, “truly transcribed.”

**Euchromatin** is **expressed**.

**DNA methylation**

Reversibly changes the expression of a DNA segment without changing its sequence. Involved with aging, carcinogenesis, epigenetics, genomic imprinting, transposable element repression, and X chromosome inactivation (lyonization).

DNA is methylated in imprinting. Methylation within gene promoter (CpG islands) typically represses (silences) gene transcription. CpG **methylation** **makes** DNA **mute**. Dysregulated DNA methylation is implicated in fragile X syndrome.

**Histone methylation**

Usually causes reversible transcriptional suppression, but can also cause activation depending on location of methyl groups.

Histone **methylation** **mostly** **makes** DNA **mute**. Lysine and arginine residues of histones can be methylated.

**Histone acetylation**

Removal of histone's  $\oplus$  charge  $\rightarrow$  relaxed DNA coiling  $\rightarrow$   $\uparrow$  transcription.

Thyroid hormone synthesis is altered by acetylation of thyroid hormone receptor. Histone **acetylation** **makes** DNA **active**.

**Histone deacetylation**

Removal of acetyl groups  $\rightarrow$  tightened DNA coiling  $\rightarrow$   $\downarrow$  transcription.

Histone deacetylation may be responsible for altered gene expression in Huntington disease. Histone **deacetylation** **deactivates** DNA.

**Nucleotides**

Nucleoside = base + (deoxy)ribose (sugar).

Nucleotide = base + (deoxy)ribose + phosphate;  
linked by 3'-5' phosphodiester bond.

Nucleo-"tri"-des have three components.

**Purines (A,G)**—2 rings.

**Pyrimidines (C,U,T)**—1 ring.

Deamination reactions:

Cytosine → uracil

Adenine → hypoxanthine

Guanine → xanthine

5-methylcytosine → thymine

Uracil found in RNA; thymine in DNA.

Methylation of uracil makes thymine.

5' end of incoming nucleotide bears the  
triphosphate (energy source for the bond).

**Pure As Gold.**

**CUT** the pyramid.

**Thymine** has a methyl.

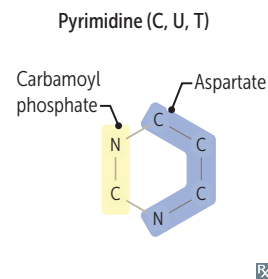
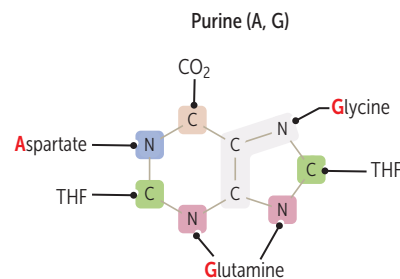
C-G bond (3 H bonds) stronger than A-T bond  
(2 H bonds). ↑ C-G content → ↑ melting  
temperature of DNA. "**C-G** bonds are like  
**Crazy G**lue."

Amino acids necessary for **purine** synthesis (cats  
**purr** until they **GAG**):

**G**lycine

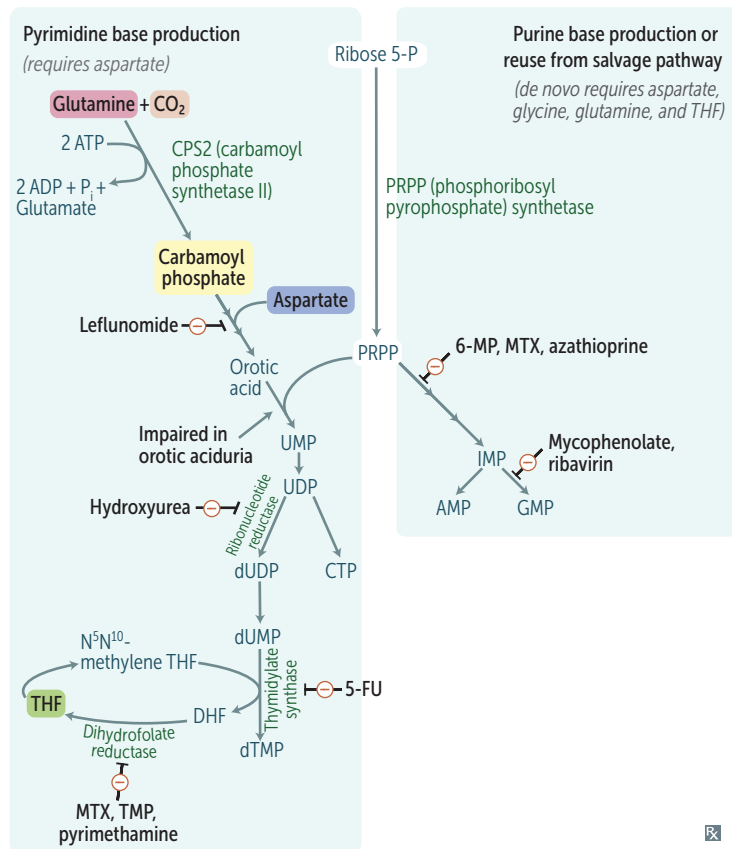
**A**spartate

**G**lutamine



### De novo pyrimidine and purine synthesis

Various immunosuppressive, antineoplastic, and antibiotic drugs function by interfering with nucleotide synthesis:



### Pyrimidine synthesis:

- **Leflunomide**: inhibits dihydroorotate dehydrogenase
- **5-fluorouracil (5-FU)** and its prodrug **capecitabine**: form 5-F-dUMP, which inhibits thymidylate synthase (↓ dTMP)

### Purine synthesis:

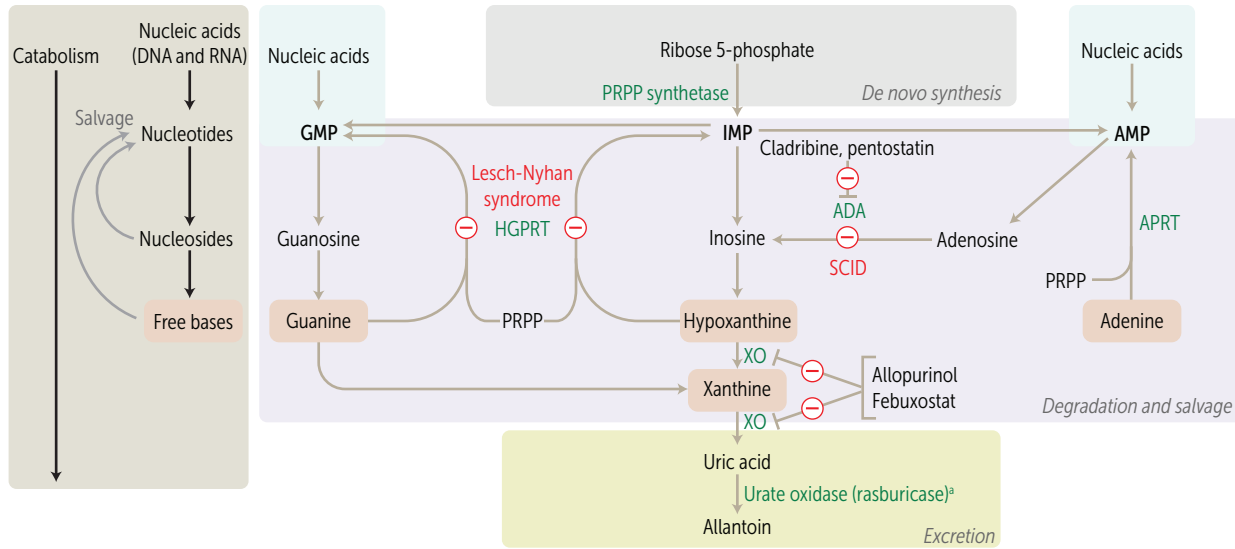
- **6-mercaptopurine (6-MP)** and its prodrug **azathioprine**: inhibit de novo purine synthesis (guanine phosphoribosyltransferase); azathioprine is metabolized via purine degradation pathway and can lead to immunosuppression when administered with xanthine oxidase inhibitor
- **Mycophenolate** and **ribavirin**: inhibit inosine monophosphate dehydrogenase

### Purine and pyrimidine synthesis:

- **Hydroxyurea**: inhibits ribonucleotide reductase
- **Methotrexate (MTX)**, **trimethoprim (TMP)**, and **pyrimethamine**: inhibit dihydrofolate reductase (↓ deoxythymidine monophosphate [dTMP]) in humans (methotrexate), bacteria (trimethoprim), and protozoa (pyrimethamine)

**CPS1** = mitochondria, urea cycle, found in liver  
**CPS2** = cytosol, pyrimidine synthesis, found in most cells

## Purine salvage deficiencies



<sup>a</sup>Absent in humans.

ADA, adenosine deaminase; APRT, adenosine phosphoribosyltransferase; HGPRT, hypoxanthine guanine phosphoribosyltransferase; XO, xanthine oxidase; SCID, severe combined immune deficiency (autosomal recessive inheritance)

## Adenosine deaminase deficiency

ADA is required for degradation of adenosine and deoxyadenosine.  $\downarrow$  ADA  $\rightarrow$   $\uparrow$  dATP  
 $\rightarrow$   $\downarrow$  ribonucleotide reductase activity  
 $\rightarrow$   $\downarrow$  DNA precursors in cells  $\rightarrow$   $\downarrow$  lymphocytes.

One of the major causes of autosomal recessive SCID.

## Lesch-Nyhan syndrome

Defective purine salvage. Deficient or mutated **HGPRT**  $\rightarrow$   $\downarrow$  GMP (from guanine) and  $\downarrow$  IMP (from hypoxanthine) formation. Compensatory  $\uparrow$  in purine synthesis ( $\uparrow$  PRPP amidotransferase activity)  $\rightarrow$  excess uric acid production. X-linked recessive. Findings: intellectual disability, self-mutilation, aggression, hyperuricemia (red/orange “sand” [sodium urate crystals] in diaper), gout, dystonia, macrocytosis.

**HGPRT:**

**H**yperuricemia  
**G**out  
**P**issed off (aggression, self-mutilation)  
**R**ed/orange crystals in urine  
**T**ense muscles (dystonia)  
 Treatment: allopurinol, febuxostat.

## Genetic code features

## Unambiguous

Each codon specifies only 1 amino acid.

Degenerate/  
redundant

Most amino acids are coded by multiple codons. **Wobble hypothesis**—first 2 nucleotides of codon are essential for anticodon recognition while the 3rd nucleotide can differ (“wobble”).

Exceptions: methionine (AUG) and tryptophan (UGG) are encoded by only 1 codon.

Commaless,  
nonoverlapping

Read from a fixed starting point as a continuous sequence of bases.

Exceptions: some viruses.

## Universal

Genetic code is conserved throughout evolution.

Exception (in animals): mitochondria (some codons specify different amino acids than in non-mitochondrial DNA).

**DNA replication**

Occurs in  $5' \rightarrow 3'$  direction (“**5**ynth**3**sis”) in continuous and discontinuous (Okazaki fragment) fashion. Semiconservative. More complex in eukaryotes than in prokaryotes, but shares analogous enzymes.

**Origin of replication A**

Particular consensus sequence in genome where DNA replication begins. May be single (prokaryotes) or multiple (eukaryotes).

AT-rich sequences (eg, TATA box regions) are found in promoters (often upstream) and origins of replication (ori).

**Replication fork B**

Y-shaped region along DNA template where leading and lagging strands are synthesized.

**Helicase C**

Unwinds DNA template at replication fork.

**Helicase halves** DNA.

Deficient in **Bloom** syndrome (**BLM** gene mutation).

**Single-stranded binding proteins D**

Prevent strands from reannealing or degradation by nucleases.

**DNA topoisomerases E**

Creates a **single-** (topoisomerase **I**) or **double-** (topoisomerase **II**) stranded break in the helix to add or remove supercoils (as needed due to underwinding or overwinding of DNA).

In eukaryotes: irinotecan/topotecan inhibit topoisomerase (TOP) I, etoposide/teniposide inhibit TOP II.

In prokaryotes: fluoroquinolones inhibit TOP II (DNA gyrase) and TOP IV.

**Primase F**

Makes RNA primer for DNA polymerase III to initiate replication.

**DNA polymerase III G**

Prokaryotes only. Elongates leading strand by adding deoxynucleotides to the  $3'$  end. Elongates lagging strand until it reaches primer of preceding fragment.

DNA polymerase III has  $5' \rightarrow 3'$  synthesis and proofreads with  $3' \rightarrow 5'$  exonuclease.

Drugs blocking DNA replication often have a modified  $3'$  OH, thereby preventing addition of the next nucleotide (“chain termination”).

**DNA polymerase I H**

Prokaryotes only. Degrades RNA primer; replaces it with DNA.

Same functions as DNA polymerase III, also excises RNA primer with  $5' \rightarrow 3'$  exonuclease.

**DNA ligase I**

Catalyzes the formation of a phosphodiester bond within a strand of double-stranded DNA.

Joins Okazaki fragments.

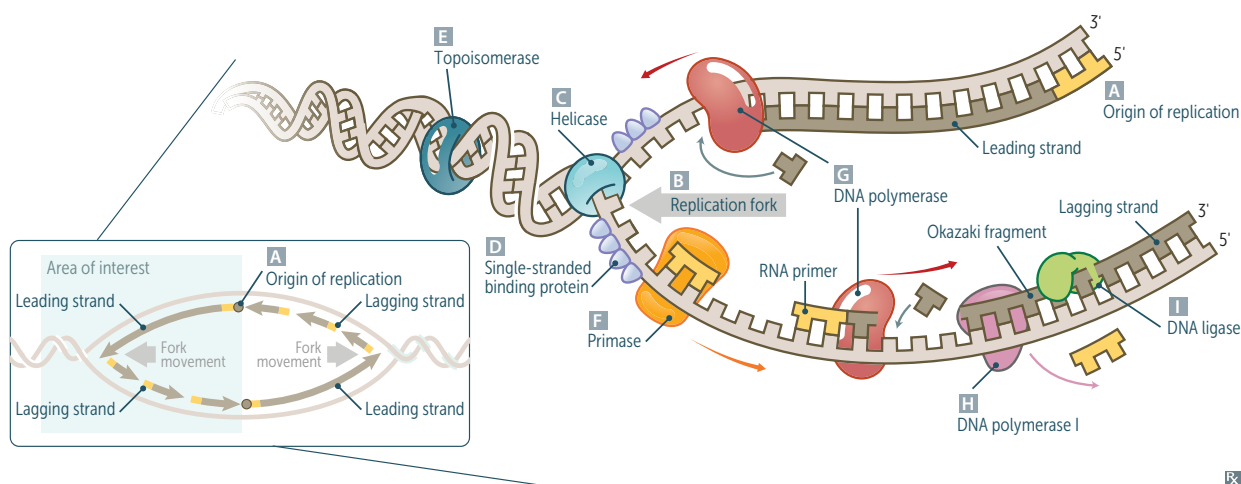
**Ligase links** DNA.

**Telomerase**

Eukaryotes only. A reverse transcriptase (RNA-dependent DNA polymerase) that adds DNA (**TTAGGG**) to  $3'$  ends of chromosomes to avoid loss of genetic material with every duplication.

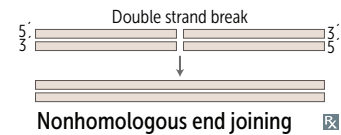
Upregulated in progenitor cells and also often in cancer; downregulated in aging and progeria.

**Telomerase TAGs** for **G**reatness and **G**lory.

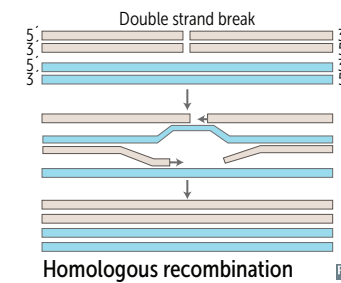


**DNA repair****Double strand****Nonhomologous end joining**

Brings together 2 ends of DNA fragments to repair double-stranded breaks. Homology not required. Part of the DNA may be lost or translocated.

**Homologous recombination**

Requires 2 homologous DNA duplexes. A strand from damaged dsDNA is repaired using a complementary strand from intact homologous dsDNA as a template. Defective in breast/ovarian cancers with *BRCA1* or *BRCA2* mutations and in certain types of Fanconi anemia. Restores duplexes accurately without loss of nucleotides.

**Single strand****Nucleotide excision repair**

Specific endonucleases remove the oligonucleotides containing damaged bases; DNA polymerase and ligase fill and reseal the gap, respectively. Repairs bulky helix-distorting lesions (eg, pyrimidine dimers).

Occurs in G<sub>1</sub> phase of cell cycle. Defective in **xeroderma pigmentosum** (inability to repair DNA pyrimidine dimers caused by UV exposure). Presents with dry skin, photosensitivity, skin cancer.

**Base excision repair**

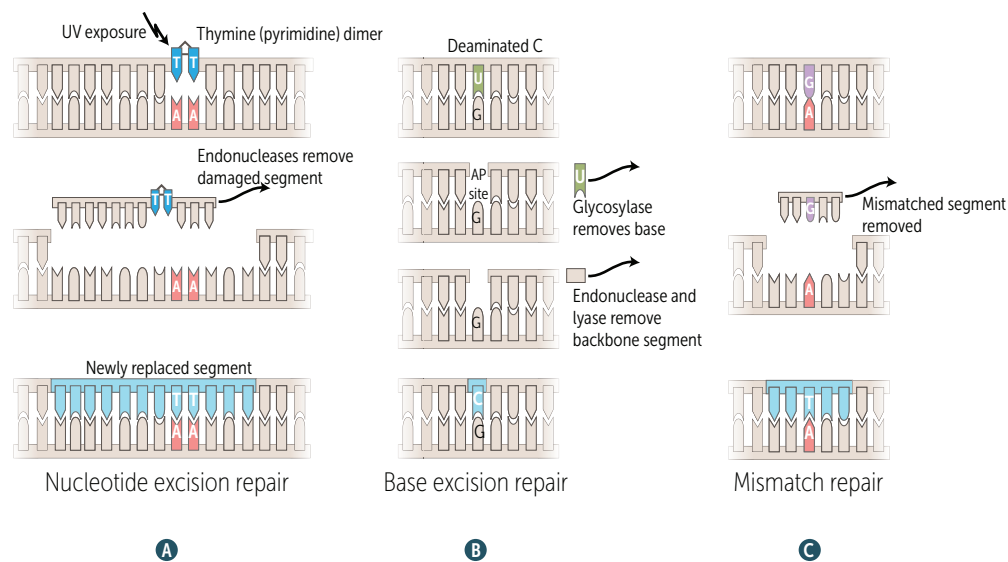
Base-specific **G**lycosylase removes altered base and creates AP (apurinic/apyrimidinic) site. AP-**E**ndonuclease cleaves 5' end, removing one or more nucleotides. AP-**L**yase cleaves 3' end. DNA **P**olymerase- $\beta$  fills the gap. DNA **L**igase seals it.

Occurs throughout cell cycle. Important in repair of spontaneous/toxic deamination. "**GEL Please**."

**Mismatch repair**

Mismatched nucleotides in newly synthesized strand are removed and gap is filled and resealed.

Occurs predominantly in S phase of cell cycle. Defective in Lynch syndrome (hereditary nonpolyposis colorectal cancer [HNPCC]).



**Mutations in DNA**

Degree of change: silent << missense < nonsense < frameshift. Single nucleotide substitutions are repaired by DNA polymerase and DNA ligase. Types of single nucleotide (point) mutations:

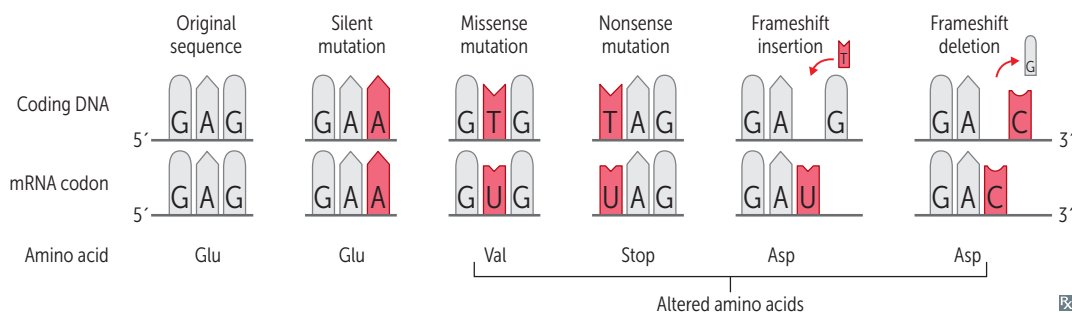
- **Transition**—purine to purine (eg, A to G) or pyrimidine to pyrimidine (eg, C to T).
- **Transversion**—purine to pyrimidine (eg, A to T) or pyrimidine to purine (eg, C to G).

**Single nucleotide substitutions**

|                          |                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Silent mutation</b>   | Codes for same (synonymous) amino acid; often involves 3rd position of codon (tRNA wobble).                                                                                      |
| <b>Missense mutation</b> | Results in changed amino acid (called conservative if new amino acid has similar chemical structure). Examples: sickle cell disease (substitution of glutamic acid with valine). |
| <b>Nonsense mutation</b> | Results in early stop codon (UGA, UAA, UAG). Usually generates nonfunctional protein. <b>Stop the nonsense!</b>                                                                  |

**Other mutations**

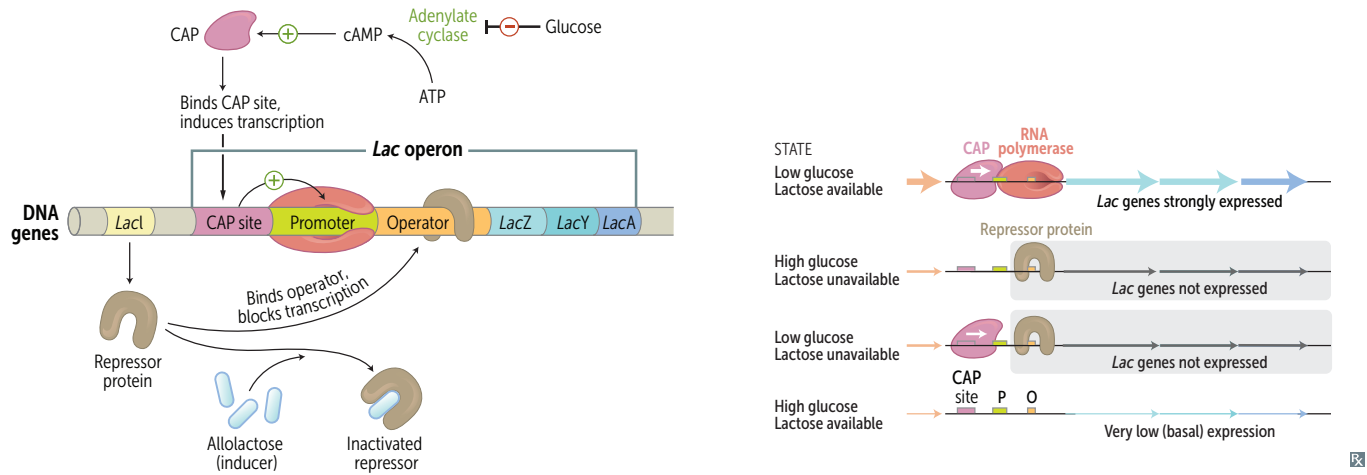
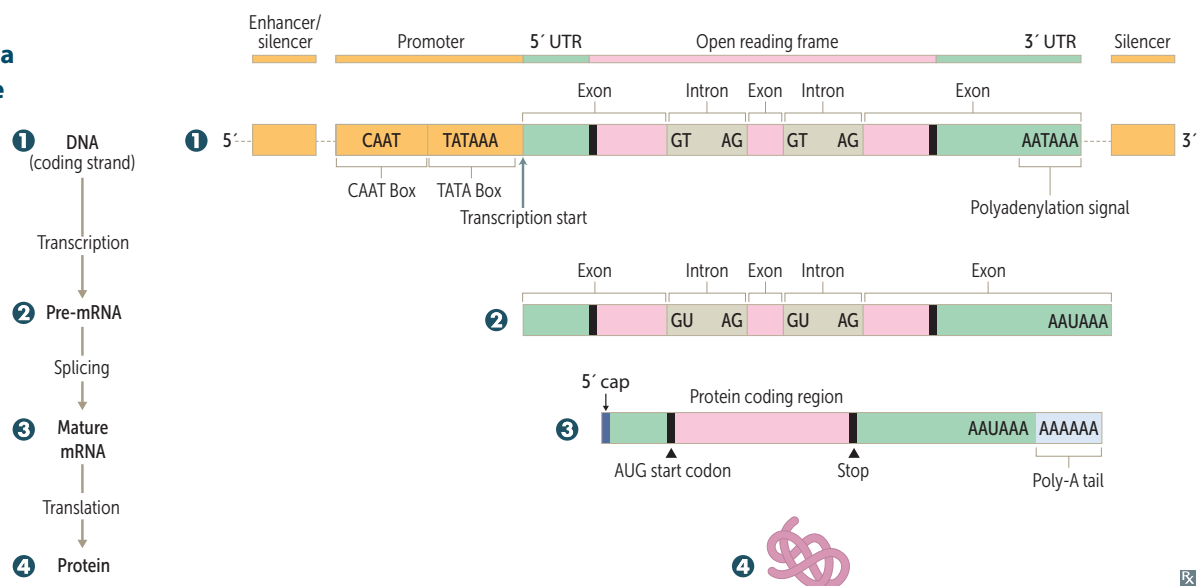
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| <b>Frameshift mutation</b>       | Deletion or insertion of any number of nucleotides not divisible by 3 (or if divisible by 3 split across adjacent codons) → misreading of all nucleotides downstream. Protein may be shorter or longer, and its function may be disrupted or altered. May occur due to slippage of DNA polymerase during replication at repetitive nucleotide regions. Examples: Duchenne muscular dystrophy, Tay-Sachs disease, cystic fibrosis. |
| <b>Splice site mutation</b>      | Retained intron in mRNA → protein with impaired or altered function. Examples: rare causes of cancers, dementia, epilepsy, some types of $\beta$ -thalassemia, Gaucher disease, Marfan syndrome.                                                                                                                                                                                                                                  |
| <b>Slipped strand mispairing</b> | Occurs when DNA polymerase slips and either inserts or removes one or more additional nucleotides by mistake in an area of repetitive nucleotides. Anticipation occurs secondary to insertion of increased repeats across generations. Example: CAG repeat expansion in Huntington disease.                                                                                                                                       |



**Lac operon**

Classic example of a genetic response to an environmental change. Glucose is the preferred metabolic substrate in *E. coli*, but when glucose is absent and lactose is available, the *lac* operon is activated to switch to lactose metabolism. Mechanism of shift:

- Low glucose → ↑ adenylate cyclase activity → ↑ generation of cAMP from ATP → activation of catabolite activator protein (CAP) → ↑ transcription.
- High lactose → unbinds repressor protein from repressor/operator site → ↑ transcription.

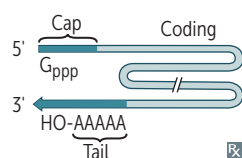
**Functional organization of a eukaryotic gene**



### Regulation of gene expression

|                    |                                                                                                                                                                                                                   |                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Promoter</b>    | Site where RNA polymerase II and multiple other transcription factors bind to DNA upstream from gene locus (AT-rich upstream sequence with TATA and CAAT boxes, which differ between eukaryotes and prokaryotes). | Promoters increase initiation of transcription. Promoter mutation commonly results in dramatic ↓ in level of gene transcription.  |
| <b>Enhancer</b>    | DNA locus where regulatory proteins (“ <b>activators</b> ”) bind, <b>increasing</b> expression of a gene on the same chromosome.                                                                                  | Enhancers and silencers may be located close to, far from, or even within (in an intron) the gene whose expression they regulate. |
| <b>Silencer</b>    | DNA locus where regulatory proteins (“ <b>repressors</b> ”) bind, <b>decreasing</b> expression of a gene on the same chromosome.                                                                                  |                                                                                                                                   |
| <b>Epigenetics</b> | Changes made to gene expression (heritable mitotically/meiotically) without a change in underlying DNA sequence.                                                                                                  | Primary mechanisms of epigenetic change include DNA methylation, histone modification, and noncoding RNA.                         |

### RNA processing (eukaryotes)



Initial transcript is called heterogeneous nuclear RNA (hnRNA). hnRNA is then modified and becomes mRNA.

The following processes occur in the nucleus:

- Capping of 5' end (addition of 7-methylguanosine cap; cotranscriptional)
- Polyadenylation of 3' end (~200 A's → poly-A tail; posttranscriptional)
- Splicing out of introns (posttranscriptional)

Capped, tailed, and spliced transcript is called mRNA.

mRNA is transported out of nucleus to be translated in cytosol.

mRNA quality control occurs at cytoplasmic processing bodies (P-bodies), which contain exonucleases, decapping enzymes, and microRNAs; mRNAs may be degraded or stored in P-bodies for future translation.

Poly-A polymerase does not require a template. AAUAAA = polyadenylation signal. Mutation in polyadenylation signal → early degradation prior to translation.

**Kozak sequence**—initiation site in most eukaryotic mRNA. Facilitates binding of small subunit of ribosome to mRNA. Mutations in sequence → impairment of initiation of translation → ↓ protein synthesis.

### RNA polymerases

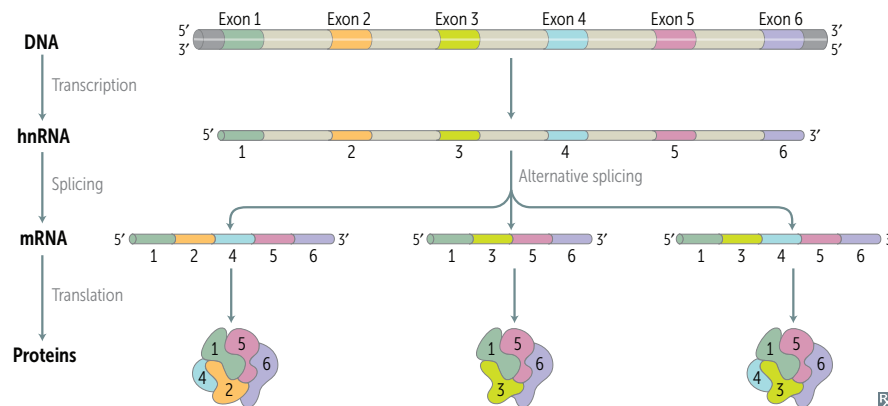
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| <b>Eukaryotes</b>  | <p>RNA polymerase I makes <b>r</b>rRNA, the most common (<b>r</b>ampant) type; present only in nucleolus.</p> <p>RNA polymerase II makes <b>m</b>mRNA (<b>m</b>assive), <b>mi</b>croRNA (<b>mi</b>RNA), and <b>s</b>mall <b>n</b>uclear RNA (<b>sn</b>RNA).</p> <p>RNA polymerase III makes 5S rRNA, <b>t</b>rRNA (<b>t</b>iny).</p> <p>No proofreading function, but can initiate chains. RNA polymerase II opens DNA at promoter site.</p> | <p>I, II, and III are numbered in the same order that their products are used in protein synthesis: rRNA, mRNA, then tRNA.</p> <p><math>\alpha</math>-amanitin, found in <i>Amanita phalloides</i> (death cap mushrooms), inhibits RNA polymerase II. Causes dysentery and severe hepatotoxicity if ingested.</p> <p>Dactinomycin inhibits RNA polymerase in both prokaryotes and eukaryotes.</p> |
| <b>Prokaryotes</b> | 1 RNA polymerase (multisubunit complex) makes all 3 kinds of RNA.                                                                                                                                                                                                                                                                                                                                                                            | Rifamycins (rifampin, rifabutin) inhibit DNA-dependent RNA polymerase in prokaryotes.                                                                                                                                                                                                                                                                                                             |

### Introns vs exons

Exons contain the actual genetic information coding for protein or functional RNA. Introns do not code for protein, but are important in regulation of gene expression. Different exons are frequently combined by alternative splicing to produce a larger number of unique proteins.

Introns are **in**tervening sequences and stay **in** the nucleus, whereas **ex**ons **ex**it and are **ex**pressed.

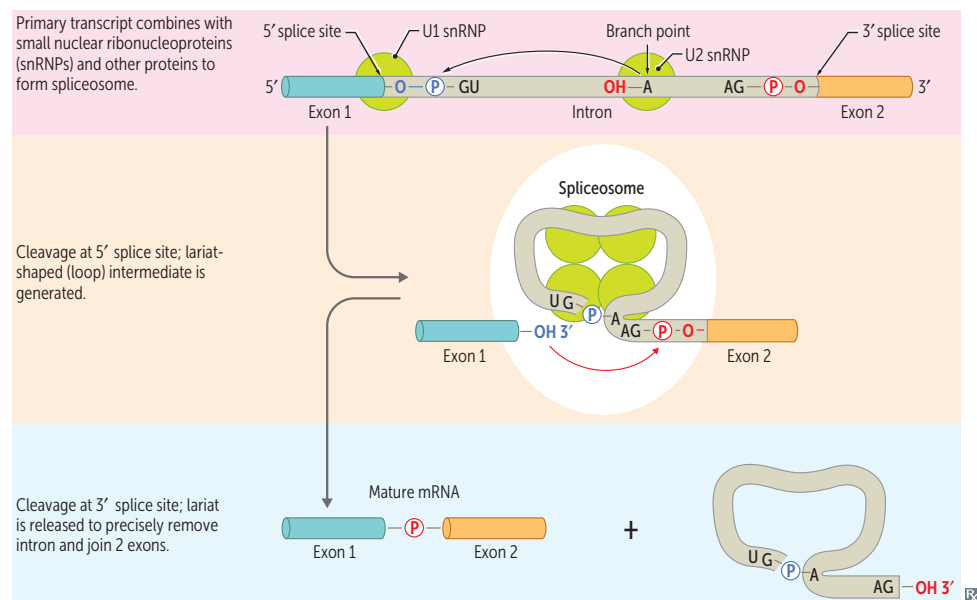
**Alternative splicing**—can produce a variety of protein products from a single hnRNA (heterogenous nuclear RNA) sequence (eg, transmembrane vs secreted Ig, tropomyosin variants in muscle, dopamine receptors in the brain, host defense evasion by tumor cells).



### Splicing of pre-mRNA

Part of process by which precursor mRNA (pre-mRNA) is transformed into mature mRNA. Introns typically begin with GU and end with AG. Alterations in snRNP assembly can cause clinical disease; eg, in spinal muscular atrophy, snRNP assembly is affected due to ↓ SMN protein → congenital degeneration of anterior horns of spinal cord → symmetric weakness (hypotonia, or “floppy baby syndrome”).

snRNPs are snRNA bound to proteins (eg, Smith [Sm]) to form a spliceosome that cleaves pre-mRNA. Anti-U1 snRNP antibodies are associated with SLE, mixed connective tissue disease, other rheumatic diseases.



**tRNA****Structure**

75–90 nucleotides, 2° structure, cloverleaf form, anticodon end is opposite 3' aminoacyl end. All tRNAs, both eukaryotic and prokaryotic, have CCA at 3' end along with a high percentage of chemically modified bases. The amino acid is covalently bound to the 3' end of the tRNA. **CCA Can Carry Amino acids.**

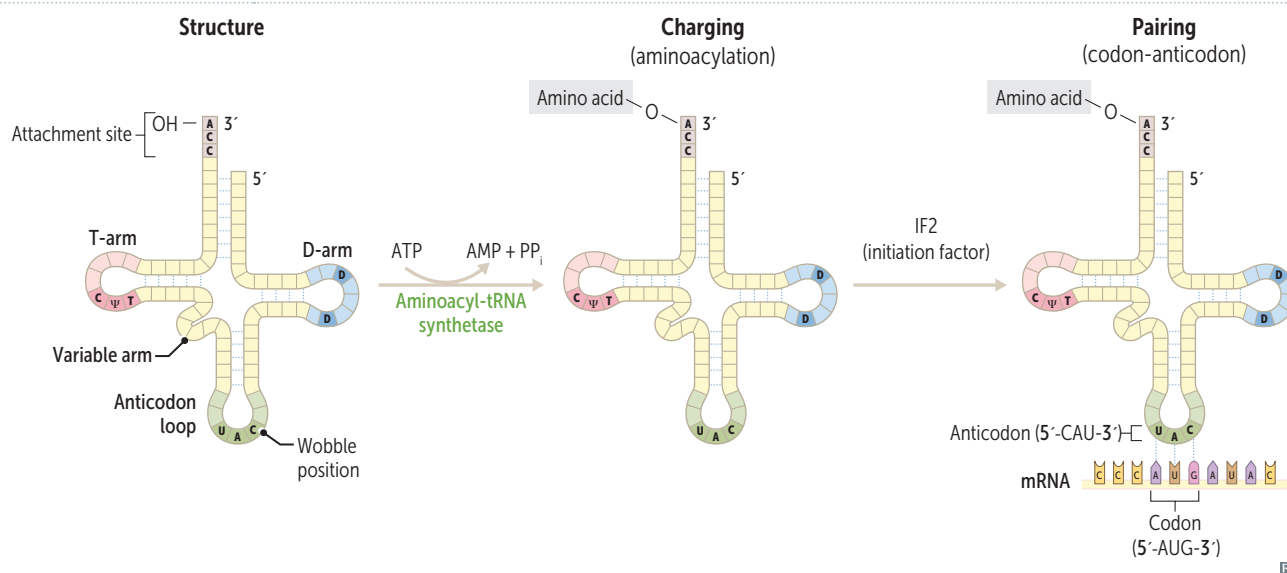
T-arm: contains the TΨC (ribothymidine, pseudouridine, cytidine) sequence necessary for tRNA-ribosome binding. **T**-arm **T**ethers tRNA molecule to ribosome.

D-arm: contains **D**ihydrouridine residues necessary for tRNA recognition by the correct aminoacyl-tRNA synthetase. **D**-arm allows **D**etection of the tRNA by aminoacyl-tRNA synthetase.

Attachment site: 3'-**ACC**-5' is the amino acid **ACC**eptor site.

**Charging**

Aminoacyl-tRNA synthetase (uses ATP; 1 unique enzyme per respective amino acid) and binding of charged tRNA to the codon are responsible for the accuracy of amino acid selection. Aminoacyl-tRNA synthetase matches an amino acid to the tRNA by scrutinizing the amino acid before and after it binds to tRNA. If an incorrect amino acid is attached, the bond is hydrolyzed. A mischarged tRNA reads the usual codon but inserts the wrong amino acid.

**Start and stop codons**

|                         |                                                                             |                                                                                                                              |
|-------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>mRNA start codon</b> | AUG.                                                                        | <b>AUG</b> in <b>AUG</b> urates protein synthesis.                                                                           |
| <b>Eukaryotes</b>       | Codes for methionine, which may be removed before translation is completed. |                                                                                                                              |
| <b>Prokaryotes</b>      | Codes for <i>N</i> -formylmethionine (fMet).                                | fMet stimulates neutrophil chemotaxis.                                                                                       |
| <b>mRNA stop codons</b> | UGA, UAA, UAG.<br>Recognized by release factors.                            | <b>UGA</b> = <b>U</b> Go <b>A</b> way.<br><b>UAA</b> = <b>U</b> Are <b>A</b> way.<br><b>UAG</b> = <b>U</b> Are <b>G</b> one. |

**Protein synthesis****Initiation**

1. Eukaryotic initiation factors (eIFs) identify the 5' cap.
2. eIFs help assemble the 40S ribosomal subunit with the initiator tRNA.
3. eIFs released when the mRNA and the ribosomal 60S subunit assemble with the complex. Requires GTP.

Eukaryotes:  $40S + 60S \rightarrow 80S$  (even).

Prokaryotes:  $30S + 50S \rightarrow 70S$  (prime).

Synthesis occurs from N-terminus to C-terminus.

ATP—tRNA **A**ctivation (charging).

GTP—tRNA **G**ripping and **G**oing places (translocation).

**Elongation**

- 1 Aminoacyl-tRNA binds to A site (except for initiator methionine, which binds the P site), requires an elongation factor and GTP.
- 2 rRNA (“ribozyme”) catalyzes peptide bond formation, transfers growing polypeptide to amino acid in A site.
- 3 Ribosome advances 3 nucleotides toward 3' end of mRNA, moving peptidyl tRNA to P site (translocation).

Think of “going **APE**”:

**A** site = incoming **A**minoacyl-tRNA.

**P** site = accommodates growing **P**eptide.

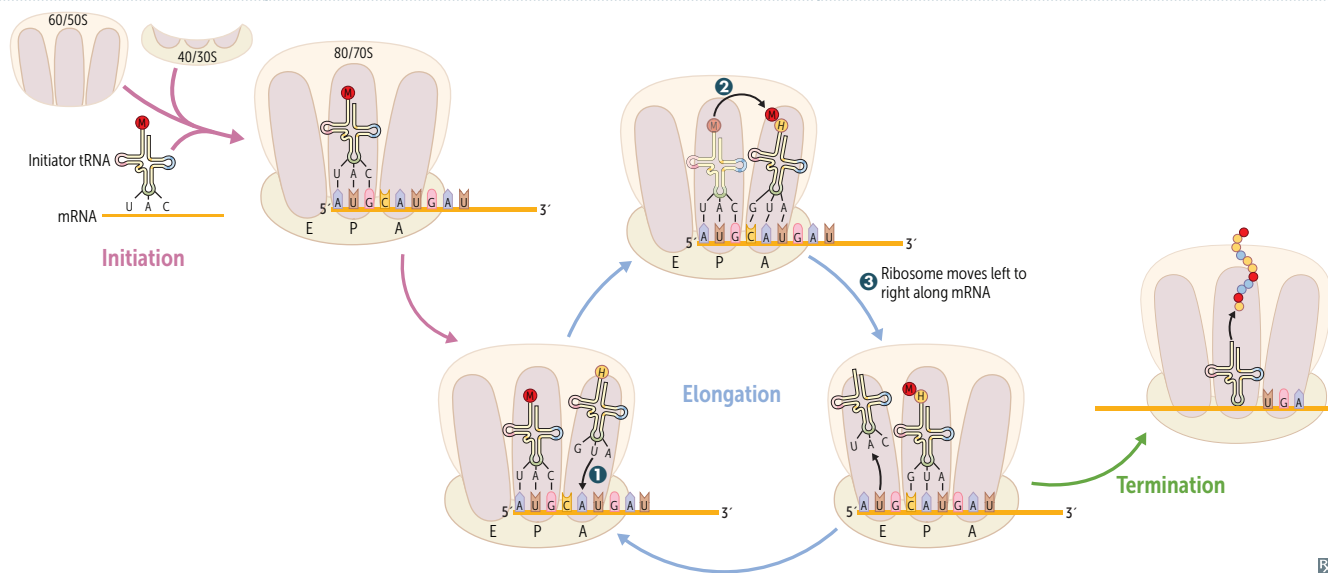
**E** site = holds **E**mpy tRNA as it **E**xits.

Elongation factors are targets of bacterial toxins (eg, *Diphtheria*, *Pseudomonas*).

**Shine-Dalgarno sequence**—ribosomal binding site in prokaryotic mRNA. Recognized by 16S RNA in ribosomal subunit. Enables protein synthesis initiation by aligning ribosome with start codon so that code is read correctly.

**Termination**

Eukaryotic release factors (eRFs) recognize the stop codon and halt translation → completed polypeptide is released from ribosome. Requires GTP.

**Posttranslational modifications****Trimming**

Removal of N- or C-terminal propeptides from zymogen to generate mature protein (eg, trypsinogen to trypsin).

**Covalent alterations**

Phosphorylation, glycosylation, hydroxylation, methylation, acetylation, and ubiquitination.

**Chaperone protein**

Intracellular protein involved in facilitating and maintaining protein folding. In yeast, heat shock proteins (eg, HSP60) are constitutively expressed, but expression may increase with high temperatures, acidic pH, and hypoxia to prevent protein denaturing/misfolding.

## ► BIOCHEMISTRY—CELLULAR

**Cell cycle phases**

Checkpoints control transitions between phases of cell cycle. This process is regulated by cyclins, cyclin-dependent kinases (CDKs), and tumor suppressors. M phase (shortest phase of cell cycle) includes mitosis (prophase, prometaphase, metaphase, anaphase, telophase) and cytokinesis (cytoplasm splits in two).  $G_1$  is of variable duration.

## REGULATION OF CELL CYCLE

**Cyclin-dependent kinases**

Constitutively expressed but inactive when not bound to cyclin.

**Cyclin-CDK complexes**

Cyclins are phase-specific regulatory proteins that activate CDKs when stimulated by growth factors. The cyclin-CDK complex can then phosphorylate other proteins (eg, Rb) to coordinate cell cycle progression. This complex must be activated/inactivated at appropriate times for cell cycle to progress.

**Tumor suppressors**

$p53 \rightarrow p21$  induction  $\rightarrow$  CDK inhibition  $\rightarrow$  Rb hypophosphorylation (activation)  $\rightarrow$   $G_1$ -S progression inhibition. Mutations in tumor suppressor genes can result in unrestrained cell division (eg, Li-Fraumeni syndrome). Growth factors (eg, insulin, PDGF, EPO, EGF) bind tyrosine kinase receptors to transition the cell from  $G_1$  to S phase.

## CELL TYPES

**Permanent**

Remain in  $G_0$ , regenerate from stem cells.

Neurons, skeletal and cardiac muscle, RBCs.

**Stable (quiescent)**

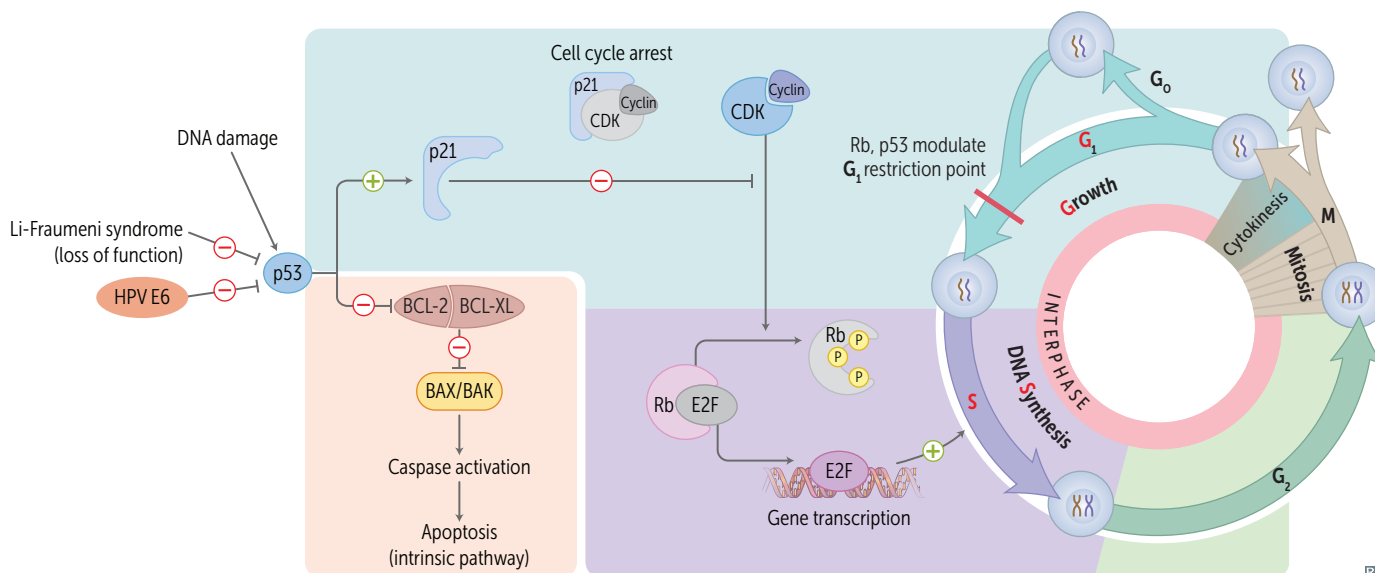
Enter  $G_1$  from  $G_0$  when stimulated.

Hepatocytes, lymphocytes, PCT, periosteal cells.

**Labile**

Never go to  $G_0$ , divide rapidly with a short  $G_1$ .  
Most affected by chemotherapy.

Bone marrow, gut epithelium, skin, hair follicles, germ cells.



**Rough endoplasmic reticulum**

Site of synthesis of secretory (exported) proteins and of N-linked oligosaccharide addition to lysosomal and other proteins.

Nissl bodies (RER in neurons)—synthesize peptide neurotransmitters for secretion.

Free ribosomes—unattached to any membrane; site of synthesis of cytosolic, peroxisomal, and mitochondrial proteins.

N-linked glycosylation occurs in the eNdoplasmic reticulum.

Mucus-secreting goblet cells of small intestine and antibody-secreting plasma cells are rich in RER.

Proteins within organelles (eg, ER, Golgi bodies, lysosomes) are formed in RER.

**Smooth endoplasmic reticulum**

Site of steroid synthesis and detoxification of drugs and poisons. Lacks surface ribosomes. Location of glucose-6-phosphatase (last step in both glycogenolysis and gluconeogenesis).

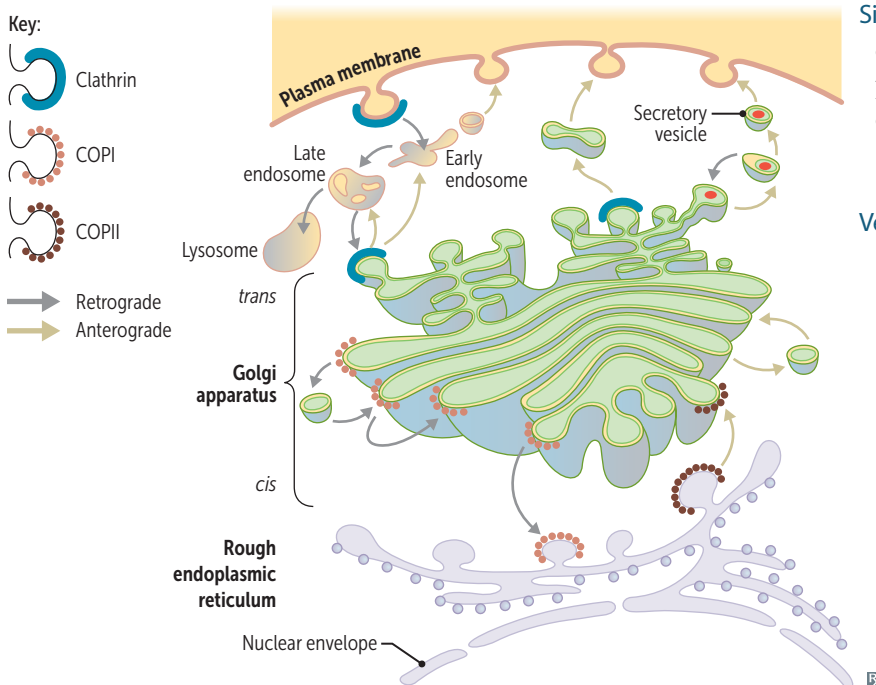
Hepatocytes and steroid hormone-producing cells of the adrenal cortex and gonads are rich in SER.

**Cell trafficking**

Golgi is distribution center for proteins and lipids from ER to vesicles and plasma membrane. Posttranslational events in GOlgi include modifying N-oligosaccharides on asparagine, adding O-oligosaccharides on serine and threonine, and adding mannose-6-phosphate to proteins for targeting to lysosomes (usually for degradation).

Endosomes are sorting centers for material from outside the cell or from the Golgi, sending it to lysosomes for destruction or back to the membrane/Golgi for further use.

**I-cell disease** (inclusion cell disease/mucopolipidosis type II)—inherited lysosomal storage disorder (autosomal recessive); defect in N-acetylglucosaminyl-1-phosphotransferase → failure of the Golgi to phosphorylate mannose residues (↓ mannose-6-phosphate) on glycoproteins → enzymes secreted extracellularly rather than delivered to lysosomes → lysosomes deficient in digestive enzymes → buildup of cellular debris in lysosomes (inclusion bodies). Results in coarse facial features, gingival hyperplasia, corneal clouding, restricted joint movements, claw hand deformities, kyphoscoliosis, and ↑ plasma levels of lysosomal enzymes. Symptoms similar to but more severe than Hurler syndrome. Often fatal in childhood.



**Signal recognition particle (SRP)**—abundant, cytosolic ribonucleoprotein that traffics polypeptide-ribosome complex from the cytosol to the RER. Absent or dysfunctional SRP → accumulation of protein in cytosol.

**Vesicular trafficking proteins**

- COPI: Golgi → Golgi (retrograde); *cis*-Golgi → ER.
- COPII: ER → *cis*-Golgi (anterograde). “**Two** (COPII) steps forward (anterograde); **one** (COPI) step back (retrograde).”
- Clathrin: *trans*-Golgi → lysosomes; plasma membrane → endosomes (receptor-mediated endocytosis [eg, LDL receptor activity]).

**Peroxisome**

Membrane-enclosed organelle involved in:

- $\beta$ -oxidation of very-long-chain fatty acids (VLCFA) (strictly peroxisomal process)
- $\alpha$ -oxidation of branched-chain fatty acids (strictly peroxisomal process)
- Catabolism of amino acids and ethanol
- Synthesis of bile acids and plasmalogens (important membrane phospholipid, especially in white matter of brain)

**Zellweger syndrome**—autosomal recessive disorder of peroxisome biogenesis due to mutated *PEX* genes (accumulation of pipecolic acid in peroxisomes). Hypotonia, seizures, jaundice, craniofacial dysmorphism, hepatomegaly, early death.

**Refsum disease**—autosomal recessive disorder of  $\alpha$ -oxidation → buildup of phytanic acid due to inability to degrade it. Vision loss (retinitis pigmentosa), anosmia, hearing loss, ataxia, peripheral neuropathy, ichthyosis, and cardiac conduction defects. Treatment: diet, plasmapheresis.

**Adrenoleukodystrophy**—X-linked recessive disorder of  $\beta$ -oxidation due to mutation in *ABCD1* gene → VLCFA buildup in **adrenal** glands, white (**leuko**) matter of brain, testes. Progressive disease that can lead to adrenal gland crisis, progressive loss of neurologic function, death.

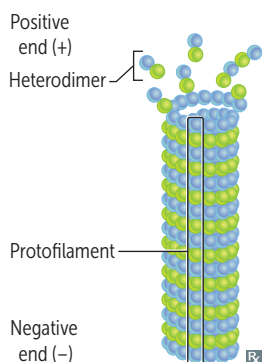
**Proteasome**

Barrel-shaped protein complex that degrades polyubiquitin-tagged proteins. Plays a role in many cellular processes, including immune response (MHC I-mediated). Defects in ubiquitin-proteasome system also implicated in diverse human diseases including neurodegenerative diseases.

**Cytoskeletal elements**

A network of protein fibers within the cytoplasm that supports cell structure, cell and organelle movement, and cell division.

| TYPE OF FILAMENT              | PREDOMINANT FUNCTION                          | EXAMPLES                                                                                      |
|-------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Microfilaments</b>         | Muscle contraction, cytokinesis, phagocytosis | Actin, microvilli                                                                             |
| <b>Intermediate filaments</b> | Maintain cell structure                       | Vimentin, desmin, cytokeratin, lamins, glial fibrillary acidic protein (GFAP), neurofilaments |
| <b>Microtubules</b>           | Movement, cell division                       | Cilia, flagella, mitotic spindle, axonal trafficking, centrioles                              |

**Microtubule**

Cylindrical outer structure composed of a helical array of polymerized heterodimers of  $\alpha$ - and  $\beta$ -tubulin. Each dimer has 2 GTP bound. Incorporated into flagella, cilia, mitotic spindles. Also involved in slow axoplasmic transport in neurons.

**Molecular motor proteins**—transport cellular cargo toward opposite ends of microtubule.

- Retrograde to microtubule (+ → -)—dynein.
- Anterograde to microtubule (- → +)—kinesin.

*Clostridium tetani* toxin, poliovirus, rabies virus, and herpes simplex virus (HSV) use dynein for retrograde transport to the neuronal cell body. HSV reactivation occurs via anterograde transport from cell body (kinesin mediated). Slow anterograde transport rate limiting step of peripheral nerve regeneration after injury.

Drugs that act on microtubules (**m**icrotubules **g**et **c**onstructed **v**ery **t**erribly):

- **M**ebendazole (anthelmintic)
- **G**riseofulvin (antifungal)
- **C**olchicine (antigout)
- **V**inca alkaloids (anticancer)
- **T**axanes (anticancer)

Negative end **n**ear **n**ucleus.

Positive end **p**oints to **p**eriphery.

**Ready? Attack!**



**Cilia structure**

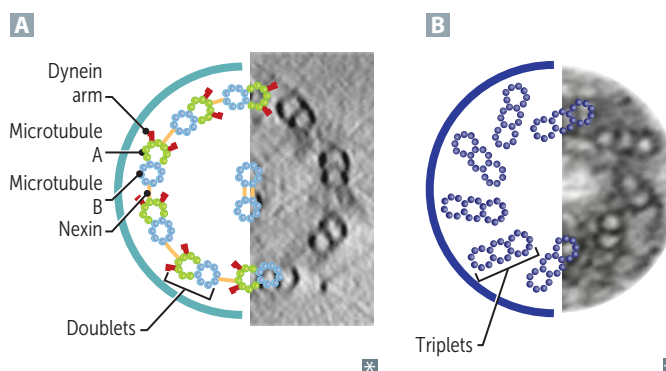
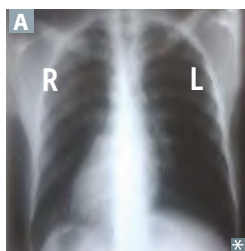
Motile cilia consist of 9 doublet + 2 singlet arrangement of microtubules (axoneme) **A**.

Basal body (base of cilium below cell membrane) consists of 9 microtubule triplets **B** with no central microtubules.

Nonmotile (primary) cilia work as chemical signal sensors and have a role in signal transduction and cell growth control. Dysgenesis may lead to polycystic kidney disease, mitral valve prolapse, or retinal degeneration.

Axonemal dynein—ATPase that links peripheral 9 doublets and causes bending of cilium by differential sliding of doublets.

Gap junctions enable coordinated ciliary movement.

**Primary ciliary dyskinesia**

Autosomal recessive. Dynein arm defect → immotile cilia → dysfunctional ciliated epithelia. Most common type is Kartagener syndrome (PCD with situs inversus).

Developmental abnormalities due to impaired migration and orientation (eg, situs inversus **A**, hearing loss due to dysfunctional eustachian tube cilia); recurrent infections (eg, sinusitis, ear infections, bronchiectasis due to impaired ciliary clearance of debris/pathogens); infertility (↑ risk of ectopic pregnancy due to dysfunctional fallopian tube cilia, immotile spermatozoa).

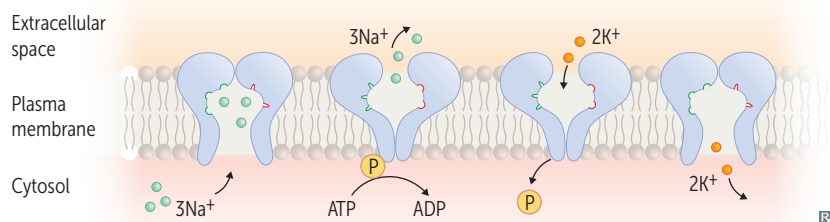
Lab findings: ↓ nasal nitric oxide (used as screening test).

**Sodium-potassium pump**

$\text{Na}^+/\text{K}^+$ -ATPase is located in the plasma membrane with ATP site on cytosolic side. For each ATP consumed, **2  $\text{K}^+$**  go **in** to the cell (pump dephosphorylated) and **3  $\text{Na}^+$**  go **out** of the cell (pump phosphorylated).

**2 strikes?  $\text{K}$ , you're still in. 3 strikes?  $\text{Na}$ h, you're out!**

Digoxin directly inhibits  $\text{Na}^+/\text{K}^+$ -ATPase → indirect inhibition of  $\text{Na}^+/\text{Ca}^{2+}$  exchange → ↑  $[\text{Ca}^{2+}]_i$  → ↑ cardiac contractility.





**Collagen**

Most abundant protein in the human body.  
Extensively modified by posttranslational modification.  
Organizes and strengthens extracellular matrix.  
Types I to IV are the most common types in humans.

Type I: **S**keleton  
Type II: **C**artilag**e**  
Type III: **A**rteries  
Type IV: **B**asement membrane  
**SCAB**

**Type I**

Most common (90%)—bone (made by osteoblasts), skin, tendon, dentin, fascia, cornea, **late** wound repair.

Type **I**: **b**one, **t**endon.  
↓ production in osteogenesis imperfecta type I.

**Type II**

Cartilage (including hyaline), vitreous body, nucleus pulposus.

Type **II**: cart**w**olage.

**Type III**

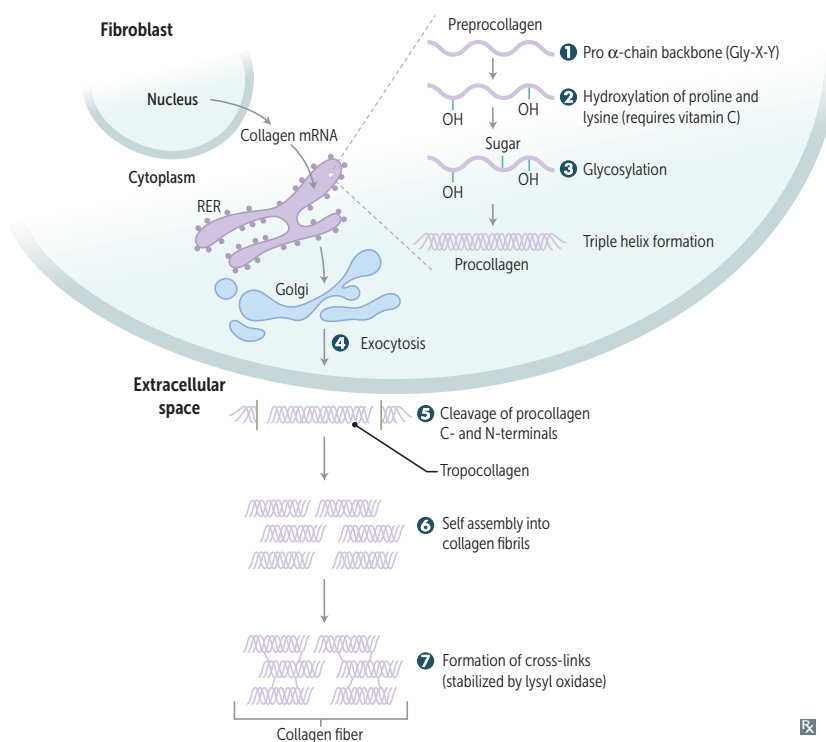
Reticulin—skin, **blood vessels**, uterus, fetal tissue, **early** wound repair.

Type **III**: deficient in **v**ascular type of **E**hlers-Danlos syndrome (**threE D**).  
Myofibroblasts are responsible for secretion (proliferative stage) and wound contraction.

**Type IV**

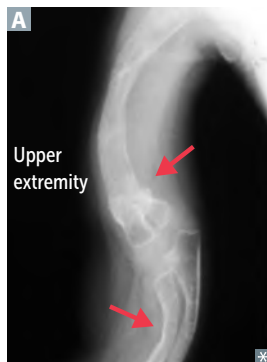
Basement membrane/basal lamina (glomerulus, cochlea), lens.

Type **IV**: under the **f**loor (basement membrane).  
Defective in Alport syndrome; targeted by autoantibodies in Goodpasture syndrome.

**Collagen synthesis and structure**

- 1 **Synthesis**—translation of collagen  $\alpha$  chains (preprocollagen)—usually Gly-X-Y (X is often proline or lysine and Y is often hydroxyproline or hydroxylysine). Collagen is 1/3 glycine; glycine content of collagen is less variable than that of lysine and proline.
- 2 **Hydroxylation**—hydroxylation (“hydrox**C**ylation”) of specific proline and lysine residues. Requires vitamin **C**; deficiency → scurvy.
- 3 **Glycosylation**—glycosylation of pro- $\alpha$ -chain hydroxylysine residues and formation of procollagen via hydrogen and disulfide bonds (triple helix of 3 collagen  $\alpha$  chains). Problems forming triple helix → osteogenesis imperfecta.
- 4 **Exocytosis**—exocytosis of procollagen into extracellular space.
- 5 **Proteolytic processing**—cleavage of disulfide-rich terminal regions of procollagen → insoluble tropocollagen.
- 6 **Assembly and alignment**—collagen assembles in fibrils and aligns for cross-linking.
- 7 **Cross-linking**—reinforcement of staggered tropocollagen molecules by **c**ovalent lysine-hydroxylysine **c**ross-linkage (by **c**opper-containing lysyl oxidase) to make **c**ollagen fibers. Cross-linking of collagen ↑ with age. Problems with cross-linking → Menkes disease.

### Osteogenesis imperfecta



Genetic bone disorder (brittle bone disease) caused by a variety of gene defects (most commonly *COL1A1* and *COL1A2*). Most common form is autosomal dominant with ↓ production of otherwise normal type I collagen (altered triple helix formation). Manifestations include:

- Multiple fractures and bone deformities (arrows in **A**) after minimal trauma (eg, during birth)
- Blue sclerae **B** due to thin, translucent scleral collagen revealing choroidal veins
- Some forms have tooth abnormalities, including opalescent teeth that wear easily due to lack of dentin (dentinogenesis imperfecta)
- Hearing loss (abnormal ossicles)

May be confused with child abuse.  
Treat with bisphosphonates to ↓ fracture risk.  
Patients can't **BITE**:

**B**ones = multiple fractures

**I** (eye) = blue sclerae

**T**eeth = dental imperfections

**E**ar = hearing loss



### Ehlers-Danlos syndrome

Faulty collagen synthesis causes skin to be hyperextensible and often thin or transparent **A**, joints to be hypermobile **B**, and tendency to bleed (easy bruising).

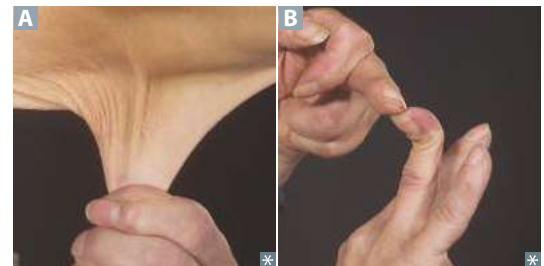
Multiple types. Inheritance and severity vary. Can be autosomal dominant or recessive. May be associated with joint dislocation, berry and aortic aneurysms, organ rupture.

Hypermobility type (joint instability): most common type.

Classical type (joint and skin symptoms): caused by a mutation in type V collagen (eg, *COL5A1*, *COL5A2*).

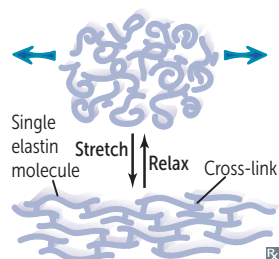
Vascular type (fragile tissues including vessels [eg, aorta], muscles, and organs that are prone to rupture [eg, gravid uterus]): mutations in type III procollagen (eg, *COL3A1*).

Can be caused by procollagen peptidase deficiency.



### Menkes disease

X-linked recessive connective tissue disease caused by impaired copper absorption and transport due to defective Menkes protein *ATP7A* (**A**bsent copper), vs *ATP7B* in Wilson disease (copper **B**uildup). Leads to ↓ activity of lysyl oxidase (copper is a necessary cofactor) → defective collagen cross-linking. Results in brittle, “kinky” hair, growth and developmental delay, hypotonia, ↑ risk of cerebral aneurysms.

**Elastin**

Stretchy protein within skin, lungs, large arteries, elastic ligaments, vocal cords, epiglottis, ligamenta flava (connect vertebrae → relaxed and stretched conformations). Rich in nonhydroxylated proline, glycine, and lysine residues, vs the hydroxylated residues of collagen.

Tropoelastin with fibrillin scaffolding.

Cross-linking occurs extracellularly via lysyl oxidase and gives elastin its elastic properties.

Broken down by elastase, which is normally inhibited by  $\alpha_1$ -antitrypsin.

$\alpha_1$ -Antitrypsin deficiency results in unopposed elastase activity, which can cause COPD.

**Marfan syndrome**—autosomal dominant (with variable expression and symptoms due to pleiotropy) connective tissue disorder affecting skeleton, heart, and eyes. **FBN1** gene mutation on chromosome 15 (fifteen) results in defective fibrillin-1, a glycoprotein that forms a sheath around elastin and sequesters TGF- $\beta$ . Findings: tall with long extremities; chest wall deformity (pectus carinatum [pigeon chest] or pectus excavatum **A**); hypermobile joints; long, tapering fingers and toes (arachnodactyly); cystic medial necrosis of aorta; aortic root aneurysm rupture or dissection (most common cause of death); mitral valve prolapse; ↑ risk of spontaneous pneumothorax.

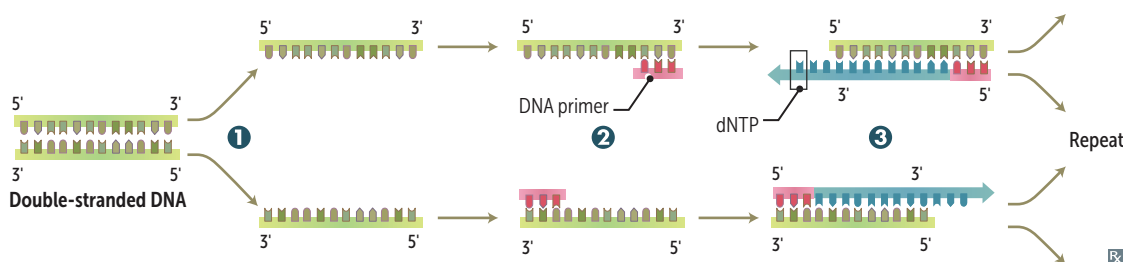
**Homocystinuria**—most commonly due to cystathionine synthase deficiency leading to homocysteine buildup. Presentation similar to Marfan syndrome with pectus deformity, tall stature, ↑ arm:height ratio, ↓ upper:lower body segment ratio, arachnodactyly, joint hyperlaxity, skin hyperelasticity, scoliosis, fair complexion (vs Marfan syndrome).

|                               | <b>Marfan syndrome</b>            | <b>Homocystinuria</b> |
|-------------------------------|-----------------------------------|-----------------------|
| <b>INHERITANCE</b>            | Autosomal dominant                | Autosomal recessive   |
| <b>INTELLECT</b>              | Normal                            | Decreased             |
| <b>VASCULAR COMPLICATIONS</b> | Aortic root dilatation            | Thrombosis            |
| <b>LENS DISLOCATION</b>       | Upward/temporal (Marfan fans out) | Downward/nasal        |

## ► BIOCHEMISTRY—LABORATORY TECHNIQUES

**Polymerase chain reaction**

Molecular biology lab procedure used to amplify a desired fragment of DNA. Useful as a diagnostic tool (eg, neonatal HIV, herpes encephalitis).



- 1 Denaturation**—DNA template, DNA primers, a heat-stable DNA polymerase, and deoxynucleotide triphosphates (dNTPs) are heated to separate the DNA strands.
- 2 Annealing**—sample is cooled. DNA primers anneal to the specific sequence to be amplified on the DNA template.
- 3 Elongation**—temperature is increased. DNA polymerase adds dNTPs to the strand to replicate the sequence after each primer.

Heating and cooling cycles continue until the amount of DNA is sufficient.